POD PDF Projection and the CMS Inclusive-Jet PDF Fit

Inclusive Jets / xFitter working note

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1 Workflow and implementation map

The calculation has two distinct layers. The direct PDF fit defines the analytic PDF parameters and experimental likelihood. The POD layer is an intermediate, deterministic map from that PDF to 100 coefficients, followed by a second evaluation of the same experimental likelihood. The coefficients are never fitted in the POD-mediated analytic fit.

The direct analytic configuration is in `central/parameters.yaml`; the evolution, reaction, data-file, and theory settings are in `central/constants.yaml` and the xFitter steering card is `central/steering.txt`. The normal direct fit is launched by `central/run.sh`. The frozen best-fit parameter values and nuisance shifts used by the fixed-nuisance comparisons are stored in `reference fit result` and `reference nuisances`; `create_reference_fit.py` generates that snapshot from a completed direct fit.

For a direct-versus-POD evaluation, `evaluate_direct.py` materialises a card directory, fixes the analytic parameters, and optionally exports the direct PDF at the POD scale and exact projection grid. The projection definition—basis, flavours, x range, scale, active modes, and the gluon/valence/F_2 closure weights—is the single `scan configuration`. `projection_metrics.py` adds the CMS-specific rows to the generic projector in `pod_projection.py`. `evaluate_pod.py` then writes the POD card and invokes xFitter. The scan driver is `run_likelihood_scan.py`, and its saved results are plotted by `plot_likelihood_scan.py`.

The POD decomposition read by xFitter is implemented in `PODPdfDecomposition.cc` and `PODBasis.cc`. The POD YAML contract is documented in `the xFitter POD README`; concrete five-mode and gluon-only seven-mode configurations are `pod_5mode/pod.yaml` and `pod_gluon_cms_7mode/pod.yaml`. The full-

POD analytic fit is orchestrated by `pod_mediated_fit/run_fit.py`, which reads the same scan configuration.

2 POD representation and projection

The POD construction follows the linear PDF-model idea of Ref. [3]. At $Q_0 = 1.65\,\mathrm{GeV}$, member zero of the LHAPDF set `250503_pod_basis_40k` is the reference PDF $f_a^{(0)}$. The 100 non-reference members define the shifts

$$B_{ai}(x,Q) = f_a^{(i)}(x,Q) - f_a^{(0)}(x,Q), \qquad i = 1,\ldots,100. \tag{1}$$

Here and below all PDF values mean xf_a. A POD PDF is therefore

$$f_a^{\mathrm{POD}}(x,Q_0;\boldsymbol{c}) = f_a^{(0)}(x,Q_0) + \sum_{i=1}^{100} B_{ai}(x,Q_0)c_i. \tag{2}$$

QCDNUM evolves the PDF represented by this input condition; the numerical coefficients c_i themselves are scale independent [5].

2.1 Design matrix

For a target PDF $t_a(x,Q_{\mathrm{ext}})$, the basis and target are always evaluated at the same Q_{ext}. In the CMS PDF-fit implementation the rows are indexed by the ordered pair (a,k), with

$$a = (g,u,\bar{u},d,\bar{d},s,\bar{s},c,\bar{c},b,\bar{b}) \tag{3}$$

and x_k the 160 retained LHAPDF nodes, $1.166\times10^{-5} \le x_k \le 1$. Thus the basic design matrix has $11\times160 = 1760$ rows and 100 columns:

$$X_{(a,k),i} = B_{ai}(x_k,Q_{\mathrm{ext}}), \qquad d_{(a,k)} = t_a(x_k,Q_{\mathrm{ext}}) - f_a^{(0)}(x_k,Q_{\mathrm{ext}}). \tag{4}$$

In other words, its ith column is the full flavour-and-x supervector of the shift member i minus member zero. All basis elements means members 1 through 100; a truncated model simply retains the requested subset of those columns. The generic CT18NNLO validation uses these native columns, with no amplitude rescaling and an absolute metric.

2.2 Minimisation

For the basic objective, the coefficients are the weighted linear least-squares solution

$$\hat{\boldsymbol{c}} = \underset{\boldsymbol{c}}{\operatorname{argmin}} \left\| W^{1/2}(X\boldsymbol{c} - \boldsymbol{d}) \right\|_2^2. \tag{5}$$

For a diagonal W, the formal normal equations are

$$(X^T W X)\hat{\boldsymbol{c}} = X^T W \boldsymbol{d}. \tag{6}$$

The CMS-specific implementation does *not* explicitly invert this normal matrix. It stacks the absolute rows and all additional weighted rows into an augmented system $A\boldsymbol{c} \simeq \boldsymbol{b}$, then calls `numpy.linalg.lstsq`, i.e. a rank-revealing LAPACK least-squares solve. There is no bound, prior, regularisation, or experimental likelihood in this step. The returned PDF is $f^{(0)} + X\hat{\boldsymbol{c}}$.

On this local machine, constructing the 1760-by-100 operator, including reading and evaluating all 101 LHAPDF members, took 6.3 s. Once built, a full CMS projection took 9 ms (average of 20 calls). The projection is therefore negligible compared with an xFitter likelihood evaluation; the operator should be built once and reused for all scan points.

3 Modifications for CMS Inclusive jets (13 TeV)

The retained CMS objective supplements the absolute residual with relative closure terms:

$$\begin{aligned}\chi^2_{\text{proj}} = \|X\boldsymbol{c} - \boldsymbol{d}\|_2^2 &+ \lambda_g^2 \sum_{0.05<x<0.99} \left(\frac{r_g}{\max(|t_g|, \epsilon)} \right)^2 \\ &+ \lambda_v^2 \sum_{10^{-4}<x<0.1} \sum_{q=u,d} \left(\frac{r_{q-\bar{q}}}{\max(|t_{q-\bar{q}}|, \epsilon)} \right)^2 \\ &+ \lambda_{F_2}^2 \sum_{10^{-4}<x<0.1} \left(\frac{r_{F_2}}{\max(|t_{F_2}|, \epsilon)} \right)^2 .\end{aligned} \tag{7}$$

Here r denotes target minus reconstruction, and

$$F_2^{\text{proxy}} = \sum_{q=u,d,s,c,b} e_q^2 (q + \bar{q}). \tag{8}$$

The implementation appends one row to A for every term in these sums, dividing both the corresponding row of X and component of d by its denominator and multiplying both by λ. Consequently, it is exactly the same

linear least-squares problem described above. The production settings are $\lambda_g = 0.1$, $\lambda_v = 0.1$, and $\lambda_{F_2} = 3$. They are projection objectives only—neither experimental likelihood terms nor PDF priors.

The relative high-x gluon term is the *gluon-tail enhancement*. An absolute norm otherwise underweights a small but phenomenologically relevant high-x gluon. The valence and F_2 terms protect low-x DIS-relevant light-quark combinations.

The direct HERAPDF input starts at $\sqrt{1.9}\,\text{GeV}$ while POD starts at 1.65 GeV. Valid closure therefore exports the direct PDF on a full LHAPDF grid down to $x = 10^{-9}$, reimports it at 1.65 GeV, and evolves both inputs through the same QCDNUM route. This prevents grid, input-scale, and evolution-history effects from being called POD compression. The implementation is in xFitter [4], with the CMS/HERA configuration supplied alongside the CMS analysis [1, 2]. The numerical values and ranges used for external projection are read from `scan_config.yaml`; the augmented-row construction itself is `projection_metrics.py`.

Pointwise high-x $u, \bar{u}, d, \bar{d}$ relative terms were rejected because vanishing antiquarks are ill conditioned. Stable high-x valence/F_2 trials also did not improve closure. The full basis spans the light combinations but only along a highly ill-conditioned direction that catastrophically damages gluon closure; these trials are diagnostics, not production settings.

4 From projected coefficients to the CMS likelihood

The projection returns only the 100-dimensional vector $\hat{\boldsymbol{c}}$. It is not itself a prediction or a fit to data. To evaluate the likelihood of the corresponding PDF, xFitter first reconstructs the complete set of input flavours at $Q_0 = 1.65\,\text{GeV}$ from the POD representation above, and evolves that input with QCDNUM at NNLO [4, 5]. The c_i are held fixed during this evolution: it is the reconstructed input PDF, rather than each coefficient separately, that is evolved. The value of $\alpha_s(m_Z)$ is passed independently of the projection, since it is not encoded in the POD coefficients.

For the CMS inclusive-jet data, the evolved PDFs and α_s are supplied to the four NNLO fastNLO interpolation tables, one per measured rapidity bin. Each table represents the NNLO partonic calculation, and the fastNLO toolkit performs the PDF and α_s convolution for the trial PDF at $\mu_R = \mu_F = p_T^{\text{jet}}$ [6]. Thus no jet cross-section variations in the POD directions are precomputed. At every coefficient vector, fastNLO obtains a new parton-level prediction

$F_i(\hat{\boldsymbol{c}})$ for each data bin. The prediction entering the measurement is

$$T_i(\hat{\boldsymbol{c}}) = F_i(\hat{\boldsymbol{c}})\, C_i^{\mathrm{NP}}\, C_i^{\mathrm{EW}}, \tag{9}$$

where the tabulated non-perturbative and electroweak correction factors are the same fixed factors as in the direct CMS fit. In parallel, the same QCDNUM-evolved PDF is passed to the neutral- and charged-current DIS reactions for the HERA data.

Schematically, xFitter evaluates the correlated experimental objective as

$$\chi^2(\hat{\boldsymbol{c}}, \boldsymbol{\beta}) = [\boldsymbol{D} - \boldsymbol{T}(\hat{\boldsymbol{c}}) - \Gamma(\hat{\boldsymbol{c}})\boldsymbol{\beta}]^T C^{-1}(\hat{\boldsymbol{c}})[\boldsymbol{D} - \boldsymbol{T}(\hat{\boldsymbol{c}}) - \Gamma(\hat{\boldsymbol{c}})\boldsymbol{\beta}] + \boldsymbol{\beta}^T\boldsymbol{\beta} + \chi^2_{\log}. \tag{10}$$

Here $\boldsymbol{D}$ denotes the measured HERA and CMS cross sections, C contains statistical and uncorrelated components (including the supplied CMS statistical correlation matrix), Γ contains the correlated experimental shifts, and β_k are unit-Gaussian Hessian nuisance parameters. The exact xFitter implementation uses the CMS analysis's Poisson statistical scaling and linear multiplicative scaling of uncorrelated and correlated uncertainties; consequently C and Γ can depend on the prediction. The final term denotes the associated logarithmic penalty. This equation is a compact description of that implemented prescription, rather than a replacement for the data cards.

For the POD-mediated analytic fit, the reported POD likelihood is the profiled value

$$\chi^2_{\mathrm{POD}}(\hat{\boldsymbol{c}}) = \min_{\boldsymbol{\beta}} \chi^2(\hat{\boldsymbol{c}}, \boldsymbol{\beta}). \tag{11}$$

Therefore the experimental nuisance shifts are reprofiled after the PDF has been projected. In the POD-mediated analytic fit, an outer optimizer varies the original 15 analytic PDF-shape parameters and α_s; each trial analytic PDF is exported at 1.65 GeV, projected to $\hat{\boldsymbol{c}}$ by the weighted least-squares construction, and assigned the profiled xFitter likelihood above. The POD coefficients are derived intermediate variables, not additional fitted parameters. The workflow is implemented by the xFitter POD decomposition module, QCDNUM evolution, the fastNLO reaction, and the projection driver in the accompanying analysis repository. In particular, `run_fit.py` calls `evaluate_direct.py`, projects the exported grid with `projection_metrics.py`, and calls `evaluate_pod.py` for the profiled POD likelihood. Its output directory stores every projected coefficient vector, input cards, console logs, likelihood result, and the outer-optimizer history.

5 Likelihood-closure plots

The closure plots test the intermediate representation without allowing a minimizer to compensate a PDF difference. Each point starts from one analytic parameter vector, evaluates the direct likelihood, projects that PDF onto the full 100-mode POD basis, and evaluates the corresponding POD likelihood. The two evaluations use identical data cards, theory settings, α_s, and stored nuisance shifts. Thus their difference is a test of the projection, POD input, and common prediction interface, not a comparison of two independently fitted PDFs.

The reference parameter vector and nuisance values are `fit_result.yaml` and `nuisances.yaml`. The scan coordinates, $[-2, 2]$ Hesse-standard-deviation range, 21 points per parameter, fixed-nuisance choice, and projection settings are in `scan_config.yaml`. `evaluate_direct.py` injects those fixed shifts through xFitter's `UseFixedNuisances` and `FixedNuisanceFile` controls. The POD counterpart, `evaluate_pod.py`, generates the immutable full-basis POD card (including $Q_0 = 1.65$ GeV, members 1–100, and all active flavours) and uses the same fixed shifts by default. Passing `-profile-nuisances` instead activates nuisance profiling; this is used for the outer POD-mediated fit, not for these closure plots.

`run_likelihood_scan.py` performs and records the paired evaluations. It first checks that the direct reference reproduces the stored fit, then writes a projection NPZ file, and will resume safely from already completed paired points. The decomposition into HERA, CMS, and HERA+CMS terms is implemented in `scan_tools.py`: each group contains its data term, its group-associated fixed-nuisance penalty, and its log penalty. Finally, `plot_likelihood_scan.py` makes one plot per analytic direction and a multipage PDF. The upper panels show both curves relative to the direct central value; the lower panels show $\chi^2_{\mathrm{POD}} - \chi^2_{\mathrm{direct}}$ directly, so a constant offset and a change of slope or minimum can be distinguished.

6 Direct CMS analytic parametrisation

At $Q_0^2 = 1.9\,\mathrm{GeV}^2$, the direct CMS PDF fit uses endpoint powers times a low-order polynomial:

$$xg = A_g x^{B_g}(1-x)^{C_g}(1 + D_g x + E_g x^2), \tag{12}$$

$$xu_v = A_{u_v} x^{B_{u_v}}(1-x)^{C_{u_v}}(1 + E_{u_v} x^2), \tag{13}$$

$$xd_v = A_{d_v} x^{B_{d_v}}(1-x)^{C_{d_v}}, \tag{14}$$

$$x\bar{U} = A_{\bar{U}} x^{B_{\bar{D}}}(1-x)^{C_{\bar{U}}}(1 + D_{\bar{U}} x), \tag{15}$$

$$x\bar{D} = A_{\bar{D}} x^{B_{\bar{D}}}(1-x)^{C_{\bar{D}}}(1 + E_{\bar{D}} x^2). \tag{16}$$

The conventions are $\bar{U} = \bar{u}$, $\bar{D} = \bar{d} + \bar{s}$, $x\bar{d} = (1-f_s)x\bar{D}$, $x\bar{s} = f_s x\bar{D}$, $f_s = 0.4$, and $A_{\bar{U}} = (1-f_s)A_{\bar{D}}$. Valence-number and momentum sum rules fix A_g, A_{u_v}, and A_{d_v}. The 16 independent parameters are

$$B_g, C_g, D_g, E_g, B_{u_v}, C_{u_v}, E_{u_v}, B_{d_v}, C_{d_v},$$
$$A_{\bar{D}}, B_{\bar{D}}, C_{\bar{D}}, E_{\bar{D}}, C_{\bar{U}}, D_{\bar{U}}, \alpha_s(m_Z).$$

7 Other global-fit parametrisations

7.1 CT18NNLO

CT18NNLO uses the CTEQ–TEA Hessian framework and low-scale endpoint powers times Bernstein-polynomial expansions [7], $xf_a = x^{a_{1,a}}(1-x)^{a_{2,a}} P_a(y)$, where y is a monotonic map such as $\sqrt{x}$. Bernstein polynomials give local shape control; sum rules and flavour relations apply at Q_0; Hessian eigenvectors encode uncertainties. CT18NNLO is the target used by the generic POD validation.

7.2 MSHT and NNPDF

MSHT uses endpoint powers times flexible Chebyshev-polynomial expansions with Hessian uncertainties [8]. NNPDF uses neural networks with preprocessing factors $xf_a = x^{\alpha_a}(1-x)^{\beta_a}\mathrm{NN}_a(x)$ and Monte-Carlo replicas [9]. All are input-scale parametrisations followed by DGLAP evolution. POD instead fixes shape functions from an ensemble and fits only linear coordinates.

8 Scope

The scale-aligned full 100-mode POD has small local likelihood differences from the direct HERA+CMS fit. This does not validate a five- or seven-mode fit; every truncation requires its own closure and minimised-fit validation.

References

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